Notes on Economics

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Chapter 1

Welcome

These notes provide an applied introduction to economics and econometrics, with an emphasis on understanding real-world economic issues and evaluating policy interventions. The material combines economic reasoning with empirical tools commonly used in public policy, health economics, and applied social science.

The focus throughout is on interpretation and judgment rather than formal derivation. Economic models and statistical methods are treated as tools for reasoning about evidence, not as ends in themselves.

1.1 Who this book is for

These notes are intended for:

- students in economics, public policy, health sciences, and related fields
- analysts and researchers working with applied economic data
- readers with basic quantitative literacy who want to understand how economic evidence is produced and interpreted

The book does not assume advanced mathematics. Familiarity with introductory statistics is helpful but not required for all chapters.

1.2 How the book is organized

The chapters progress from concepts to methods to applications:

- Early sections introduce economic thinking, scarcity, incentives, institutions, and market structure.
- Middle sections cover core microeconomic and macroeconomic ideas, including policy tools.
- Later sections focus on empirical methods, including regression, time series analysis, and quasi-experimental designs used for policy evaluation.

Chapters are designed to be read independently, but later material builds on ideas introduced earlier.

1.3 Using figures and visuals

Visual figures are used throughout the book to anchor intuition before formal models or code are introduced. Figures are intended to:

- clarify relationships between economic actors and variables
- illustrate dynamic patterns such as trends, cycles, and interventions
- support interpretation of statistical results

Figures are referenced directly in the text and can be revisited as summaries of key ideas.

1.4 Using R code

Some sections include R code to illustrate empirical workflows.

- Code chunks are included as examples and templates.
- Some chunks may be set with eval = FALSE so the book can render without requiring all packages or datasets.
- Readers who wish to run the code locally can enable evaluation after installing required packages and ensuring data availability.

The emphasis is on understanding the logic of analysis rather than reproducing results exactly.

1.5 Learning goals

By the end of these notes, readers should be able to:

- explain core economic concepts and relate them to real-world policy questions
- interpret common economic indicators such as GDP, inflation, and unemployment
- understand the logic and limitations of regression-based analysis
- recognize issues of trend, autocorrelation, and stationarity in time-series data
- assess policy interventions using quasi-experimental reasoning, including interrupted time series analysis

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1.6 Notes on scope and interpretation

These notes emphasize interpretation, assumptions, and context. Economic evidence is always produced within institutional, data, and modeling constraints. Throughout the book, attention is given to limitations and uncertainty alongside results.

Readers are encouraged to treat the material as a foundation for critical analysis rather than as a set of mechanical procedures.

Chapter 2

What is Economics?

Economics studies how people and organizations make choices under scarcity. Scarcity implies trade-offs, and those trade-offs are shaped by institutions, incentives, information, and power. This chapter builds a shared vocabulary for the rest of the book.

Roadmap

This chapter moves from definitions to values, then to economic actors and markets. It ends by connecting market structure and market failure to why policy evaluation matters.

Learning objectives

- Define scarcity, opportunity cost, and incentives.
- Explain why economic evaluation involves values such as efficiency and fairness.
- Compare broad traditions of economic thought and their information assumptions.
- Identify key actors and describe how their objectives can differ.
- Recognize common market structures and what they imply for power.

Figure 2.1 is a high-level map. It is not a model with numbers; it is a way to organize thinking. In applied work, disagreements often come from different assumptions about which arrows are strong, which are weak, and which are constrained by institutions.

2.1 Scarcity and opportunity cost

Scarcity means that resources (time, money, labour, land, attention, public budgets) are limited relative to what people want or need. Opportunity cost

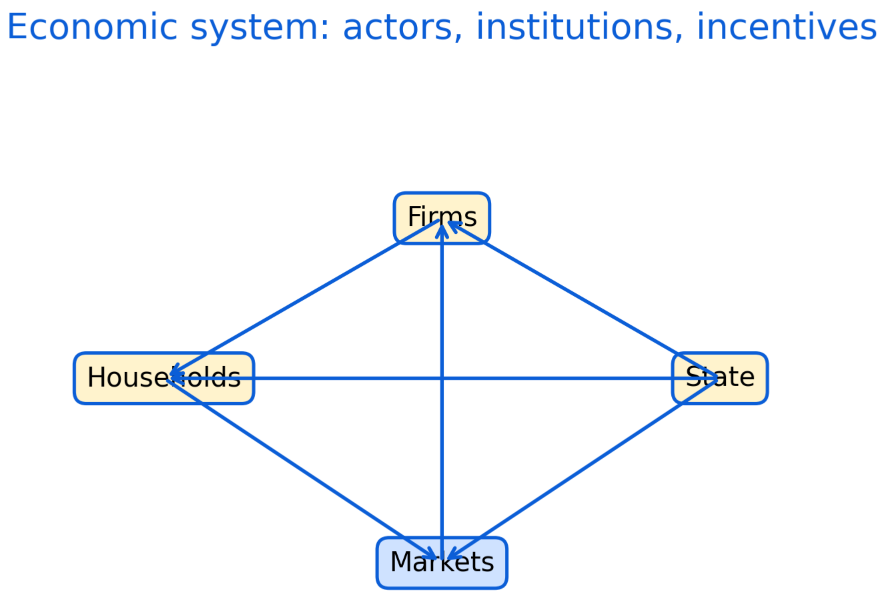


Figure 2.1: Economic system overview showing households, firms, markets, and the state. Use this map as a guide for where policy and evidence enter the system.

captures the idea that choosing one option usually means giving up another. In policy contexts, opportunity costs can be non-monetary, such as foregone services, worse health outcomes, or environmental degradation.

In applied analysis, opportunity cost is often the difference between a world with an intervention and a world without it. That is why counterfactual reasoning is central later in the book.

2.2 Economics as evidence and as judgment

Economics uses data and models to explain patterns, but it also uses criteria to judge outcomes.

Efficiency is one criterion, but it comes in more than one form. Narrow efficiency focuses on maximizing outputs for given inputs. Broader efficiency can incorporate system waste, spillovers, and long-run sustainability.

Fairness, trust, freedom, and equality appear when policies affect different groups differently. In practice, many debates are really about which values are prioritized and what trade-offs are acceptable.

2.3 Economic theories and assumptions

Different traditions emphasize different mechanisms. Social and institutional perspectives highlight that markets are embedded in social norms and legal rules. Post-Keynesian approaches emphasize uncertainty, market power, and effective demand. Neoclassical approaches often start from idealized markets and then add frictions.

These traditions differ in how they treat information. Closed-information assumptions make cost-benefit analysis easier but can omit hard-to-measure impacts. Open-information approaches encourage wider evidence but can be harder to summarize.

2.4 Economic actors

Households, firms, the state, and community actors have different objectives. Firms may pursue profit, market share, stability, or stakeholder goals. States can pursue growth, stability, redistribution, and provision of public goods. Community actors can stabilize local systems through cooperation, mutual aid, and governance.

Households are rarely a single decision-maker. Household bargaining, risk pooling, caregiving, and division of labour can affect labour supply, consumption, and health outcomes.

2.5 Markets and market structure

Market structure shapes prices, entry, innovation, and power. Monopoly and monopsony concentrate power on one side of the market. Oligopoly creates strategic interaction among large sellers. Monopolistic competition is common in differentiated goods and services. Perfect competition is a benchmark that helps organize reasoning, even if it is rarely fully met.

2.6 Market failure and why it matters

Market failures include market power, information problems, externalities, public goods, and missing markets. Many health and social policy questions arise because markets do not naturally deliver equitable or efficient outcomes.

Common pitfalls

- Treating efficiency as the only objective and ignoring distribution.
- Assuming markets are competitive when market power is present.
- Using a narrow cost-benefit frame when important impacts are unmeasured.
- Forgetting that households and communities are actors, not background.

Key takeaways

- Economics is about choices under scarcity and the trade-offs those choices create.
- Values and institutions shape what is measured and what counts as success.
- Market structure is a practical way to think about power and policy needs.

Chapter 3

Microeconomics Essentials

Microeconomics explains how prices and quantities emerge, how incentives shape behaviour, and how policies change outcomes. This chapter focuses on tools that are used repeatedly in applied policy analysis.

Roadmap

We start with supply and demand, then build intuition for elasticity and incidence. We then connect market structure to market power and end with externalities and public goods as key reasons for intervention.

Learning objectives

- Use supply and demand to explain equilibrium and comparative statics.
- Interpret elasticity and connect it to behavioural response.
- Explain incidence and why burdens do not always fall where policies are applied.
- Describe market power and practical regulatory tools.
- Explain externalities and public goods and why markets may underprovide them.

Figure 3.1 highlights why economists often talk about more than just equilibrium. Distribution of gains and losses is central for policy.

3.1 Supply, demand, and equilibrium

Demand summarizes how quantity demanded changes with price, holding other factors constant. Supply summarizes how quantity supplied changes with price. Equilibrium is the price and quantity where supply equals demand.

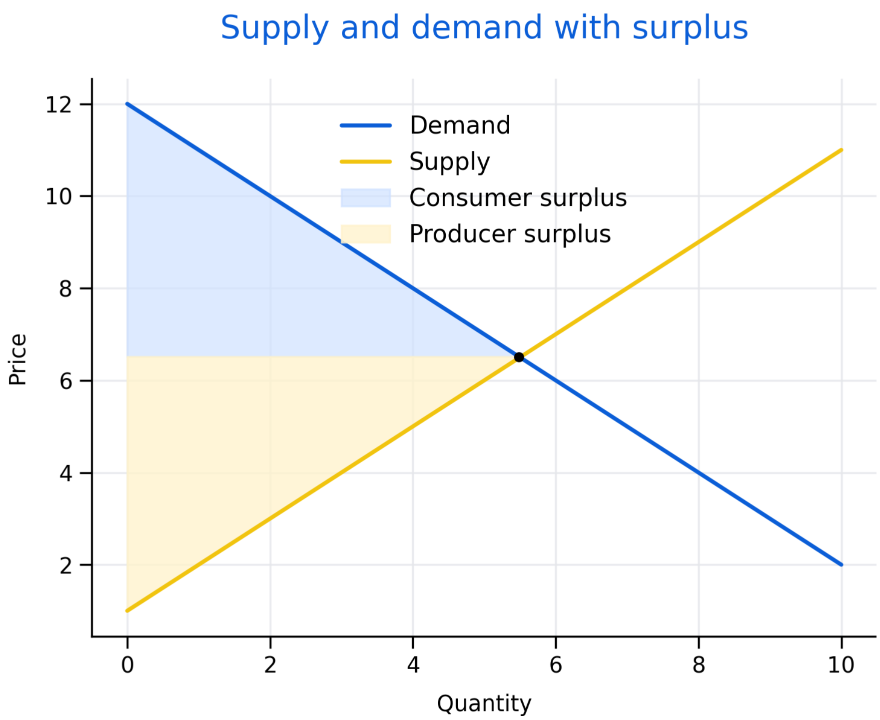


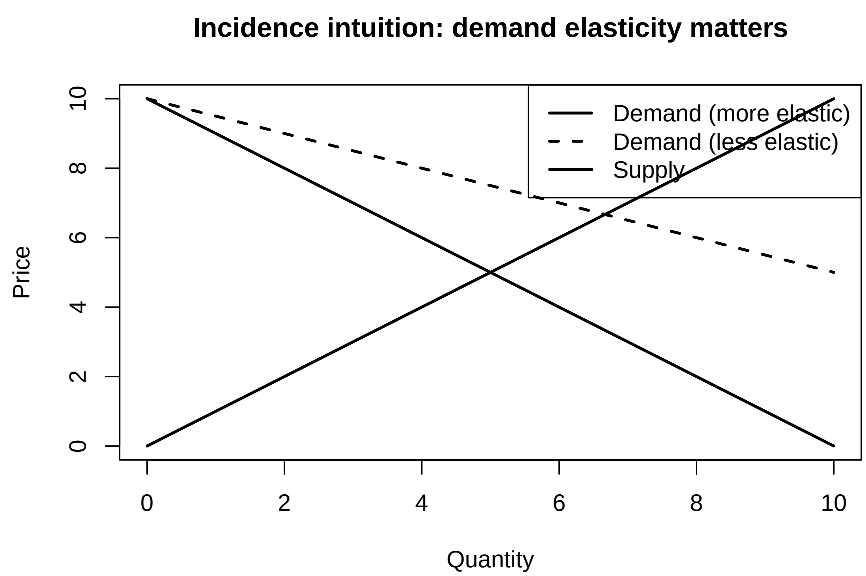
Figure 3.1: Supply and demand with welfare areas. The shaded regions illustrate how policies can shift not only prices and quantities but also consumer and producer surplus.

Comparative statics studies how equilibrium changes when curves shift. Income, preferences, and substitutes can shift demand. Input costs, technology, and regulation can shift supply.

3.2 Elasticity and incidence

Elasticity measures responsiveness. Price elasticity of demand is the percent change in quantity demanded for a one percent change in price. Supply elasticity is defined similarly.

Elasticity matters because it shapes incidence. When demand is relatively inelastic, consumers bear more of a tax burden. When supply is relatively inelastic, producers bear more. This is a key lesson for policy design: legal incidence and economic incidence are different.



3.3 Market power in practice

Market power arises when firms or buyers can influence prices. Barriers to entry, networks, scale economies, and control over inputs all contribute.

Policy tools include antitrust, regulation, procurement design, transparency requirements, and benchmarking. In public services, market design choices (who can bid, how contracts are structured, how quality is measured) can be as important as formal regulation.

3.4 Externalities and public goods

Externalities occur when actions impose costs or benefits on others that are not reflected in market prices. Public goods are non-rival and non-excludable. Both motivate intervention because private decisions do not reflect social costs and benefits.

Common policy approaches include taxes, subsidies, standards, tradable permits, and direct public provision. The right tool depends on measurement feasibility, enforcement, and equity goals.

Common pitfalls

- Treating elasticity as a fixed number rather than context-dependent.
- Assuming a tax always raises revenue or always reduces consumption substantially.
- Ignoring quality, access, and enforcement when discussing market interventions.
- Forgetting that market power can exist on the buyer side (monopsony).

Key takeaways

- Equilibrium is a starting point; welfare and incidence matter for policy.
- Elasticity shapes behavioural response and who bears burdens.
- Market power and externalities are central reasons for regulation.

Chapter 4

Macroeconomics: Measurement and Trends

Macroeconomics studies the economy as a whole. It focuses on measurement, growth, inflation, labour markets, and cycles. This chapter emphasizes what indicators mean and what they miss.

Roadmap

We introduce GDP, inflation, and unemployment measurement. We then discuss growth versus cycles and explain why distribution and inequality matter for interpreting macro indicators.

Learning objectives

- Describe what GDP measures and what it misses.
- Explain how inflation is measured and why indices can differ.
- Interpret unemployment and participation and understand limitations.
- Distinguish long-run growth from short-run business cycles.
- Explain why distribution affects macro interpretation and policy.

Figure 4.1 is a reminder that macro conditions rarely move in one direction. Stabilization policy often manages trade-offs.

4.1 GDP and national accounts

GDP is the value of final goods and services produced within a country over a period. It is useful as a measure of activity, but it is not a measure of welfare.

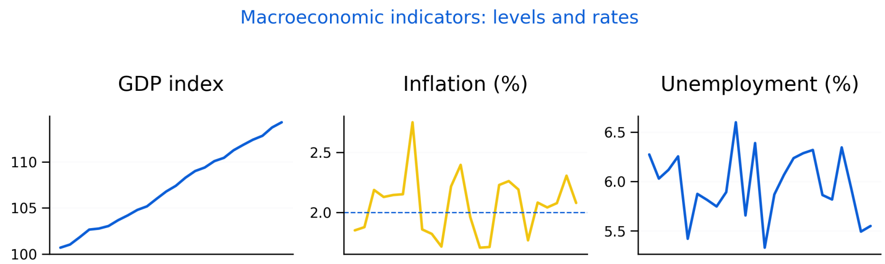


Figure 4.1: Illustrative dashboard of macro indicators. Multi-dimensional indicators help distinguish growth, price stability, and labour market conditions.

GDP can miss unpaid care work, distribution, environmental costs, and quality change. It can also rise due to rebuilding after shocks, complicating interpretation.

4.2 Inflation and price indices

Inflation measures average price changes. The CPI is common, but results depend on the basket, substitution, quality adjustment, and treatment of housing.

Inflation is not just a statistic; it affects purchasing power, wages, and distribution. High inflation can shift resources unpredictably, while disinflation can be associated with unemployment depending on conditions.

4.3 Unemployment, participation, and labour markets

Unemployment depends on definitions of job search and labour force participation. Participation can change due to demographics, discouragement, caregiving, and education.

Complementary measures include underemployment, long-term unemployment, and employment-to-population ratios.

4.4 Growth and cycles

Growth reflects long-run productivity, capital accumulation, technology, and human capital. Business cycles are fluctuations around trend caused by demand shocks, supply shocks, financial conditions, and expectations.

4.5 Distribution, poverty, and inequality

Distribution matters because aggregate indicators can hide divergent experiences across groups and regions. Poverty measures depend on thresholds and cost-of-living. Inequality can be summarized using percentiles, ratios, and indices.

Common pitfalls

- Treating GDP growth as equivalent to improved well-being.
- Interpreting inflation without considering what is driving it.
- Using unemployment alone without participation and underemployment.
- Ignoring distribution when summarizing macro performance.

Key takeaways

- Macro indicators are useful but incomplete; interpretation requires context.
- Growth and cycles have different causes and policy implications.
- Distributional analysis strengthens macroeconomic assessment.

Chapter 5

Fiscal and Monetary Policy

Fiscal and monetary policy are key tools for stabilization and long-run goals. Policies affect output, employment, inflation, and distribution, often with time lags and uncertainty.

Roadmap

We review fiscal instruments, then monetary policy and transmission. We close with trade-offs, credibility, and practical limits.

Learning objectives

- Describe core fiscal policy instruments and channels.
- Explain deficits and debt in relation to stabilization and sustainability.
- Describe how central banks influence interest rates and expectations.
- Explain why policy works with lags and uncertainty.
- Discuss trade-offs and distributional consequences.

Figure 5.1 provides a simple map of mechanisms. In practice, the strength of each channel depends on institutions, expectations, and financial conditions.

5.1 Fiscal policy

Fiscal policy includes government spending, taxation, and transfers. Spending affects demand directly. Taxes and transfers affect disposable income and incentives.

Automatic stabilizers such as progressive taxation and unemployment insurance dampen cycles without new legislation. Discretionary policy refers to deliberate changes in spending or taxes.

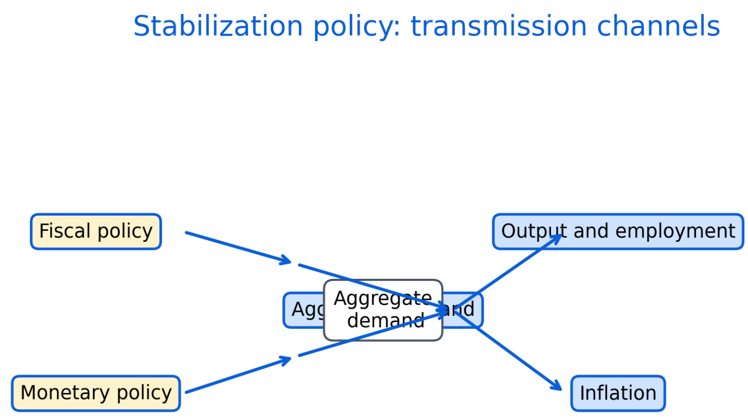


Figure 5.1: Transmission channels from fiscal and monetary policy to aggregate demand, output, employment, and inflation. Channels operate with lags and may differ by context.

Debt sustainability depends on the relationship between interest rates and growth, the credibility of fiscal institutions, and the capacity to raise revenue.

5.2 Monetary policy

Monetary policy is conducted by central banks, often with a mandate related to inflation and economic stability. Central banks influence short-term interest rates and broader financial conditions.

Transmission channels include borrowing costs, asset prices, exchange rates, and expectations. Policy lags mean effects can arrive after conditions have changed.

Financial stability concerns can require additional tools such as macroprudential regulation.

5.3 Practical limits and trade-offs

Policies face uncertainty, political constraints, and measurement challenges. Trade-offs include inflation control versus employment stabilization and distributional effects of interest rates and taxation.

Common pitfalls

- Treating policy effects as immediate rather than lagged.
- Ignoring distributional impacts of stabilization tools.
- Assuming one-size-fits-all transmission across countries or regions.
- Confusing short-run stabilization goals with long-run structural reforms.

Key takeaways

- Fiscal and monetary policy work through multiple channels with lags.
- Sustainability depends on context, institutions, and interest-growth dynamics.
- Good evaluation considers both effectiveness and equity.

Chapter 6

Ordinary Least Squares (OLS): Simple Linear Regression

OLS is a baseline method for estimating linear relationships and forming predictions. It is widely used because it is interpretable, computationally simple, and often a reasonable first approximation.

Roadmap

We explain what OLS estimates and how to interpret coefficients. We then discuss assumptions, common output statistics, and diagnostics. We end with practical responses when assumptions fail.

Learning objectives

- Interpret coefficients in a simple regression as conditional associations.
- Explain residuals and why least squares minimizes squared errors.
- Recognize key assumptions and why violations affect inference.
- Interpret common output such as R-squared, t-tests, and F-tests.
- Identify diagnostics for autocorrelation and non-constant variance.

Figure 6.1 highlights a key lesson: interpretation depends on both the relationship and the leftover structure in residuals.

The code below shows what a baseline OLS workflow looks like in practice. It is a template (not meant to run as-is) and uses robust standard errors to reflect common real-world data issues.

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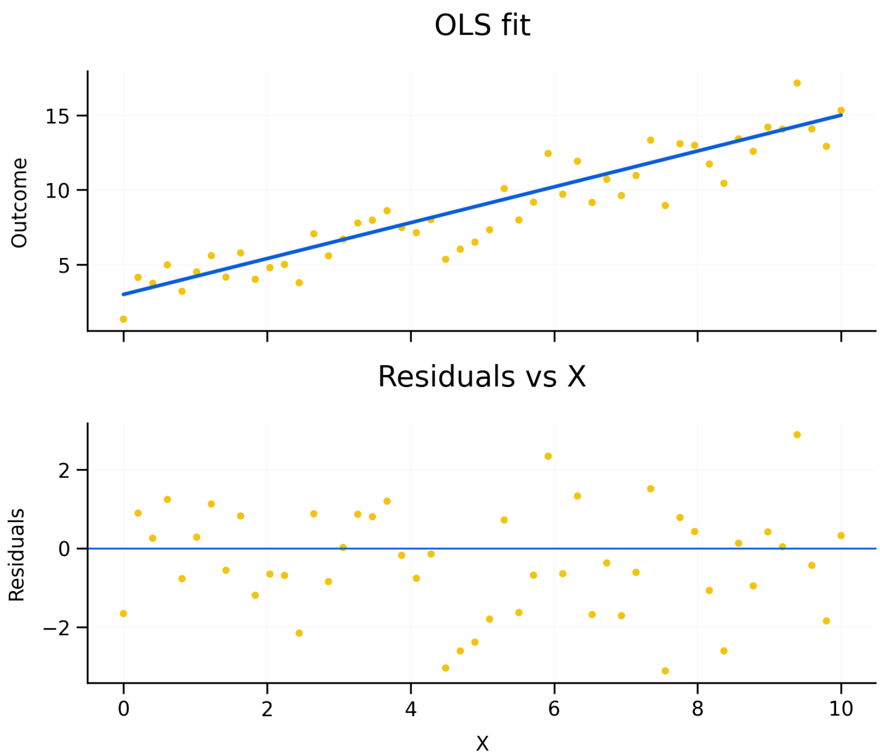

Figure 6.1: OLS fit and residual diagnostics. A good fit does not guarantee valid inference; residual patterns can indicate missing variables, non-linearity, or time-series dependence.

```r
# Example data (simulated) - replace this block with your real series when available
set.seed(1)
y <- 100 + cumsum(rnorm(60, 0, 2)) + 10*sin(2*pi*(1:60)/12)

# Example data (simulated) - replace with your real dataset
df <- data.frame(
  outcome = rnorm(100),
  exposure = rbinom(100, 1, 0.5),
  age = sample(18:90, 100, TRUE),
  sex = sample(c("F","M"), 100, TRUE)
)

# Template: OLS as a baseline model
fit <- lm(outcome ~ exposure + age + sex, data = df)
summary(fit)

# Optional: robust standard errors
# install.packages(c("sandwich", "lmtest"))
library(sandwich)
library(lmtest)
coeftest(fit, vcov. = vcovHC(fit, type = "HC1"))
```

6.1 What OLS estimates

In a simple linear regression, the coefficient summarizes average change in the outcome when the predictor changes by one unit, holding other modeled factors fixed.

OLS chooses coefficients to minimize the sum of squared residuals. Residuals capture what the model does not explain.

6.2 Assumptions and why they matter

Homoskedasticity implies constant error variance. When variance changes, standard errors can be wrong.

Zero conditional mean is the core assumption for causal interpretation. Violations often reflect omitted variables, simultaneity, or measurement error.

Independence matters for cross-sectional sampling. In time series, serial correlation is common and must be addressed.

6.3 Reading output

R-squared describes the share of variation explained. It does not measure causal validity.

t-tests evaluate whether a coefficient differs from zero given estimated uncertainty. The F-test evaluates joint significance.

AIC and BIC support model comparison by penalizing complexity.

6.4 Diagnostics and practical fixes

When assumptions fail, options include robust standard errors, adding relevant controls, transforming variables, or using models suited to the data structure.

Common pitfalls

- Interpreting coefficients causally without an identification strategy.
- Using R-squared as a measure of truth rather than fit.
- Ignoring heteroskedasticity and serial correlation in inference.
- Overfitting with many predictors and then over-interpreting significance.

Key takeaways

- OLS is useful but inference depends on assumptions.
- Diagnostics should guide whether corrections or alternative models are needed.
- Causal claims require design, not just regression.

Chapter 7

Multivariate Modeling

Applied questions often involve multiple predictors and sometimes multiple outcomes. Multivariate modeling generalizes regression thinking and raises important inference issues such as multiple testing and joint interpretation.

Roadmap

We clarify terminology, then discuss interpretation when many predictors or outcomes are present. We close with practical guidance on specification and communication.

Learning objectives

- Distinguish multiple regression from multivariate multiple regression.
- Explain why joint inference matters when outcomes are multiple.
- Describe the role of controls and interaction terms.
- Recognize risks of multiple comparisons and overfitting.
- Communicate multivariate results clearly for policy audiences.

Figure 7.1 illustrates why multivariate work requires careful interpretation. Differences across outcomes can reflect mechanisms, measurement, or specification choices.

7.1 Multiple regression

Multiple regression uses multiple predictors to explain one outcome. Controls can reduce confounding when they capture relevant omitted factors.

However, controls can also introduce bias if they are affected by the treatment (post-treatment variables). In evaluation contexts, the causal graph matters.

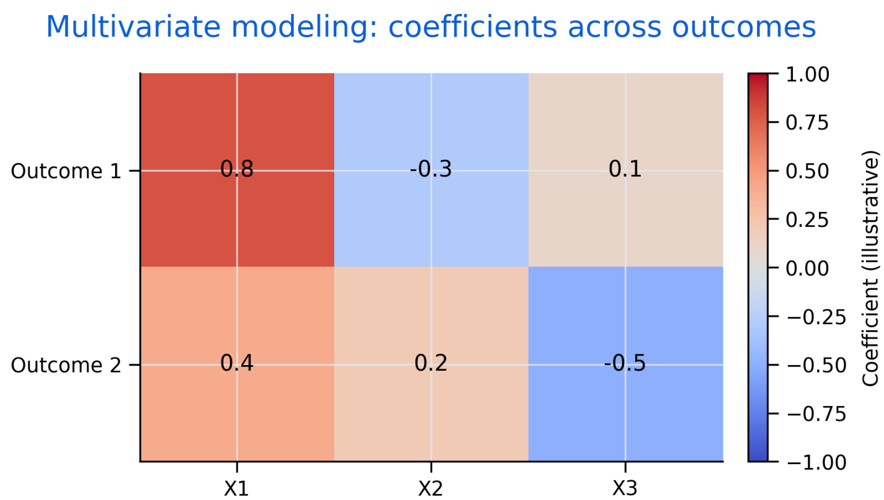

Figure 7.1: Illustrative coefficient map across outcomes. The same predictor can relate differently to different outcomes, motivating joint interpretation and careful reporting.

7.2 Multivariate multiple regression

Multivariate multiple regression models multiple outcomes using the same predictors. It is useful when outcomes are related and when you want a coherent story across measures.

7.3 Inference with many outcomes

Testing many outcomes increases false positives. Strategies include focusing on primary outcomes, using joint tests, and applying corrections.

7.4 Practical reporting

Good reporting states:

- which outcomes were primary
- how multiple testing was handled
- whether results are robust across specifications
- how effects translate into meaningful units

Common pitfalls

- Treating a long list of significant results as strong evidence without correction.
- Using controls mechanically without considering causal structure.
- Reporting only significant outcomes and omitting the full set.

Key takeaways

- Multivariate models can strengthen evidence when designed carefully.
- Joint inference and transparent reporting prevent misleading conclusions.

Chapter 8

Time Series Forecasting Theory (AR, MA, ARMA, ARIMA)

Time series data record values over time. Forecasting requires understanding persistence, trend, seasonality, and whether the series is stable. Many economic series are non-stationary, so transformations are often needed.

Roadmap

We introduce time series components, define stationarity, and explain AR and MA intuition. We then describe ARIMA notation and a practical Box-Jenkins workflow.

Learning objectives

- Identify trend, seasonality, cycles, and noise in a series.
- Explain stationarity and why differencing can help.
- Describe AR and MA intuition and how ACF/PACF guide orders.
- Explain ARIMA(p,d,q) and seasonal extensions.
- Outline a workflow for building and diagnosing forecasting models.

Figure 8.1 shows that a series can have structure beyond random noise. Forecasting aims to model that structure without overfitting it.

8.1 Components and visualization

Trend is a long-run movement. Seasonality repeats at fixed intervals. Cycles are longer fluctuations that may not repeat regularly. Noise is unpredictable

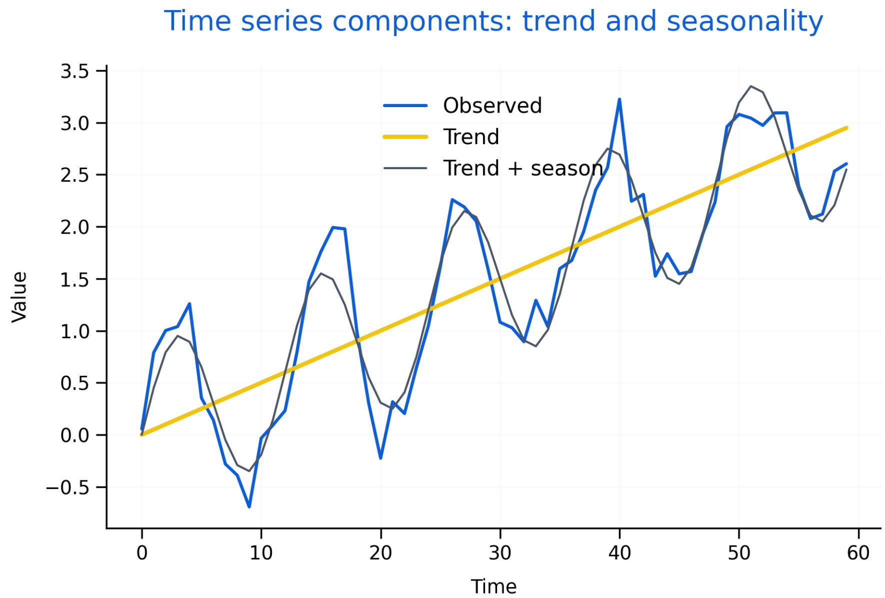

Figure 8.1: Observed time series decomposed into trend and seasonality (illustrative). Decomposition supports model selection and clarifies why differencing or seasonal adjustment may be needed.

variation.

Visualization is the first diagnostic step. A series that looks like white noise is not forecastable beyond its mean.

8.2 Stationarity and differencing

Stationarity means statistical properties do not change over time. Many economic series have trends, changing variance, or evolving autocorrelation.

Differencing removes trends and can help achieve stationarity. Seasonal differencing can remove repeating patterns.

The code below illustrates a minimal time-series workflow: plot the series, difference it if trend is present, then (optionally) run a stationarity test. It is written as a template so the book can render even if you do not have data or packages installed.

```
# Template: check for non-stationarity and difference a series
# y should be a numeric vector or ts object (e.g., monthly counts)
y_ts <- ts(y, frequency = 12)
```

```r
# Quick visual check
plot(y_ts, main = "Time series (raw)")

# If trend is present, try differencing
dy <- diff(y_ts)
plot(dy, main = "First difference")

# Formal test (requires tseries)
# install.packages("tseries") #if needed
library(tseries)
adf.test(y_ts)
adf.test(na.omit(dy))
```

8.3 AR, MA, and ARIMA intuition

AR models use past values. MA models use past shocks. ARMA combines both. ARIMA adds differencing when the original series is non-stationary.

ACF and PACF help propose orders. Information criteria help compare models. Residual checks test whether structure remains.

Common pitfalls

- Fitting ARIMA to non-stationary data without differencing.
- Choosing high orders that fit noise rather than signal.
- Ignoring seasonality when it is visually present.
- Treating forecasts as certainty rather than probabilistic ranges.

Key takeaways

- Stationarity is central for many time-series models.
- ACF/PACF, information criteria, and residual checks form a practical workflow.
- Forecasts should be evaluated for plausibility and uncertainty.

Chapter 9

Causality, Identification, and Policy Evaluation

Policy analysis often asks causal questions: what is the effect of a policy, program, or exposure on an outcome? Answering this requires a credible counterfactual.

Roadmap

We explain why correlation is not causation, then outline common identification strategies and a checklist for evaluating evidence.

Learning objectives

- Define counterfactual reasoning and why it is central.
- Identify major threats to causal interpretation.
- Describe common identification strategies and when they are plausible.
- Use a checklist to assess credibility and robustness.

Figure 9.1 summarizes major strategies. The right choice depends on what variation exists and what assumptions are defensible.

9.1 Threats to causal inference

Correlation can reflect reverse causality, omitted variables, selection, and measurement error. In applied work, these threats are often present simultaneously.

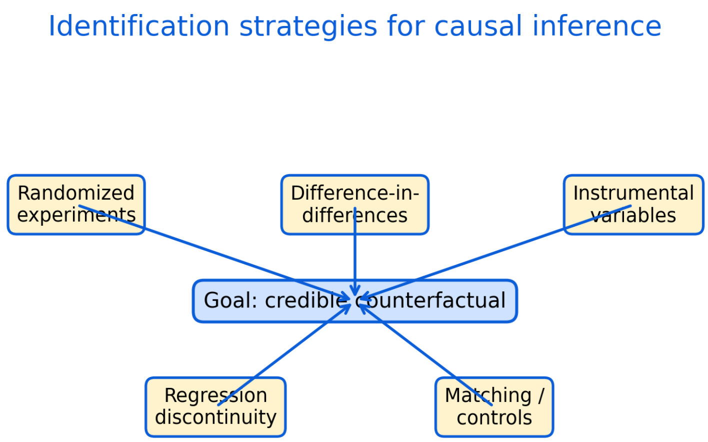

Figure 9.1: Identification strategies for causal inference. Different designs create different approximations to the counterfactual; credibility depends on assumptions and context.

9.2 Common strategies

Randomized experiments use assignment to balance factors.

Difference-in-differences uses changes over time between treated and comparison groups.

Instrumental variables use a variable that shifts treatment but affects outcomes only through treatment.

Regression discontinuity uses thresholds to create local comparisons.

Matching and controls attempt to compare similar units but rely on strong assumptions.

9.3 Practical checklist

A credible evaluation should state the causal question, define the counterfactual, explain assumptions, test robustness, and interpret uncertainty.

Common pitfalls

- Treating statistical significance as proof of causality.
- Using controls without considering causal pathways.

- Ignoring policy implementation details that break design assumptions.

Key takeaways

- Causal inference depends on design, not only models.
- A clear counterfactual and transparent assumptions improve credibility.

Chapter 10

Interrupted Time Series Analysis (ITSA): Concepts

Interrupted time series analysis evaluates interventions when randomization is not feasible. It uses the pre-intervention trend as a counterfactual and examines whether the post-intervention series shows an immediate level change, a slope change, or both.

Roadmap

We define ITSA, describe segmented regression components, then discuss assumptions and common threats. We close with extensions such as covariates, seasonality, and comparison series.

Learning objectives

- Define ITSA and explain when it is appropriate.
- Interpret level and slope changes as policy effects.
- Identify assumptions and threats such as co-interventions.
- Understand why autocorrelation requires special handling.
- Describe extensions such as controlled ITSA.

Figure 10.1 distinguishes two common intervention effects. Some policies cause immediate discontinuities; others change trends over time.

The code below shows the standard segmented-regression structure behind ITSA: a baseline trend, an intervention indicator (level change), and a post-intervention trend term (slope change).

```
# Template: segmented regression for ITSA
# t0 = the time index where the intervention starts (e.g., month 25)
```

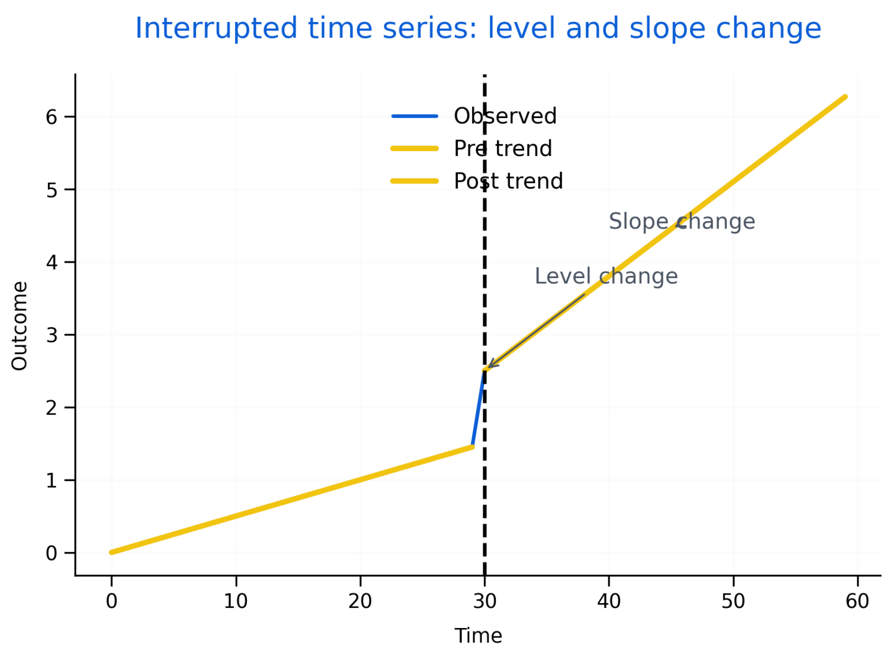

Figure 10.1: Interrupted time series showing an intervention point with both a level change and a slope change. These components map directly to segmented regression parameters.

```r
# Example data (simulated) - replace this block with your real dataset
set.seed(1)
n <- 60
t0 <- 31  # intervention starts at time point 31 (adjust as needed)

df <- data.frame(
  time = 1:n
)

# Create a simple outcome with a baseline trend + level change + slope change
df$intervention <- ifelse(df$time >= t0, 1, 0)
df$post_time <- pmax(0, df$time - t0 + 1)

df$outcome <- 10 + 0.10*df$time + 2*df$intervention + 0.05*df$post_time + rnorm(n, 0, 0.5)

# Fit segmented regression
model_itsa <- lm(outcome ~ time + intervention + post_time, data = df)
summary(model_itsa)

# Interpretation:
# - intervention = immediate level change at t0
# - post_time    = change in slope after t0
```

10.1 Segmented regression interpretation

A basic ITSA model estimates baseline trend, an immediate shift at the intervention, and a change in slope afterward. Interpretation depends on implementation timing and theory of change.

10.2 Assumptions and threats

The central assumption is that the pre-intervention trend would have continued absent the intervention. Threats include co-interventions, changes in measurement, seasonality, and regression to the mean.

Autocorrelation is common in time series and can invalidate naive standard errors.

10.3 Extensions

Extensions include covariates, interaction terms for heterogeneity, seasonality controls, time-series error structures, and comparison series.

Common pitfalls

- Declaring a causal effect without checking co-interventions.
- Using too few pre-intervention points to assess trend.
- Ignoring autocorrelation and seasonality.

Key takeaways

- ITSA is powerful when trends are stable and intervention timing is clear.
- Diagnostics and robustness checks are essential for credible claims.

Chapter 11

Interrupted Time Series Analysis (ITSA): Worked Workflow

This chapter summarizes a practical workflow for implementing ITSA. The goal is to structure applied work so that assumptions are checked, diagnostics are reported, and results are communicated clearly.

Roadmap

We outline a step-by-step process from data preparation to model refinement and reporting. The workflow is compatible with regression-based ITSA and time-series error models.

Learning objectives

- Prepare and structure time-series data for ITSA.
- Use plots to check trends, outliers, and seasonality.
- Specify a segmented regression model and interpret parameters.
- Diagnose autocorrelation and adjust inference.
- Report results with uncertainty and sensitivity checks.

Figure 11.1 is a checklist for applied work. It helps ensure that evaluation is transparent and reproducible.

After fitting a segmented regression, a key check is whether residuals are autocorrelated.

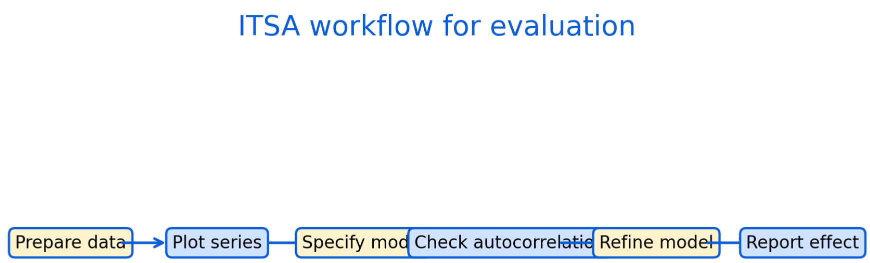

Figure 11.1: Practical ITSA workflow: prepare data, visualize, specify model, check autocorrelation, refine, and report effects with uncertainty and sensitivity checks.

```
# Template: basic diagnostics after a segmented regression
# install.packages(c("lmtest","sandwich","forecast")) if needed

library(lmtest)
library(forecast)

# model_itsa is the fitted model from the segmented regression
dwtest(model_itsa)                    # Durbin-Watson test for autocorrelation
checkresiduals(model_itsa)            # residual plot + ACF check
```

11.1 Reporting recommendations

A strong ITSA report includes:

- clear definition of the intervention and timing
- justification for pre and post windows
- plots of the outcome with intervention markers
- model specification with level and slope terms
- autocorrelation diagnostics and corrected inference
- sensitivity analysis (alternative windows, functional forms, seasonality)

Common pitfalls

- Reporting a single model with no sensitivity checks.
- Omitting diagnostic evidence for autocorrelation.

- Over-interpreting short post-intervention windows.

Key takeaways

- A transparent workflow improves credibility.
- Sensitivity checks are part of good evaluation practice.